



Network Group Assignment (Core)

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Network Group Assignment Case Study (Core)



Learning Objectives:

- Given a scenario, develop a threat model and countermeasures.

Scenario:

You work for a PR and marketing company that handles highly sensitive information for its high-profile clients. Client records are stored in a database and file system hosted on your private corporate network. As well as client records, this includes media such as photos and videos. Most remote client communications and data transfers take place using a one-to-one encrypted messaging app, but you also accommodate some clients who prefer to use email. A high percentage of your staff work remotely, accessing data and services over a VPN. You are reviewing your security procedures in light of some high-profile hacks of celebrity data. At this point, you want to understand the attack surface and attack vectors by which your private network could be compromised.



Questions:

1. What remote access methods could an attacker exploit?
2. Focusing on email, think of how email is processed as it is sent by a remote user and received by your company. Give 1 possible attack vector for each level of adversary capability.
3. What comes next in the chain of processing incoming email, and what attack vectors can adversaries exploit?
4. What countermeasures can be deployed for each email attack vector?

Assignment Submission

Given the scenario above, answer the 4 questions given. Work on this as a team but submit the document individually for grading/tracking purposes. The names of all participants should be on the document.

Assignment Solution

What remote access methods could an attacker exploit?

1. Remote Access Methods:

Possible Attack Vectors:

- **Exploiting VPN vulnerabilities:** Attackers could target vulnerabilities in the VPN infrastructure to gain unauthorized access to the corporate network.
- **Phishing attacks on remote employees:** Using social engineering, attackers might trick employees into revealing VPN credentials or downloading malicious files.

Countermeasures:

- Regularly update and patch VPN software to address known vulnerabilities.
- Conduct employee training on recognizing and avoiding phishing attacks.

Focusing on email, think of how email is processed as it is sent by a remote user and received by your company. Give 1 possible attack vector for each level of adversary capability.

2. Email Processing:

Email Sent by a Remote User:

- **Adversary with Low Capability:**
 - **Attack Vector:** Spoofing sender addresses to impersonate a trusted source.
 - **Countermeasure:** Implement SPF, DKIM, and DMARC to authenticate and validate sender identities.
- **Adversary with Medium Capability:**
 - **Attack Vector:** Sending malicious attachments or links in emails.
 - **Countermeasure:** Use advanced threat protection (ATP) to scan and filter attachments and links for malicious content.
- **Adversary with High Capability:**
 - **Attack Vector:** Crafting sophisticated spear-phishing emails tailored to individual employees.

- **Countermeasure:** Implement user awareness training to educate employees about spear-phishing techniques.

3. Chain of Processing Incoming Email:

Adversaries can exploit:

- **Email Filtering Systems:** Attackers might employ evasion techniques to bypass traditional email filters.
- **Mail Servers:** Target vulnerabilities in mail server software to gain unauthorized access.
- **User Mailboxes:** Exploit weak passwords or conduct credential stuffing attacks.

Countermeasures:

- Deploy advanced email filtering systems that use machine learning to detect sophisticated phishing attempts.
- Regularly update and patch mail server software.
- Enforce strong password policies and implement multi-factor authentication (MFA) for user mailboxes.

4. Countermeasures for Email Attack Vectors:

- **Spoofed Sender Addresses:**
 - Implement SPF, DKIM, and DMARC.
 - Train employees to verify email sender information.
- **Malicious Attachments/Links:**
 - Use advanced threat protection (ATP) to scan and filter attachments and links.
 - Encourage employees to verify the legitimacy of unexpected attachments or links.
- **Spear-Phishing:**
 - Provide ongoing security awareness training.
 - Implement email authentication mechanisms and conduct regular phishing simulations.

- **Email Filtering Systems:**
 - Invest in advanced email filtering solutions that use behavioral analysis and machine learning.
 - Regularly update and monitor filtering rules.
- **Mail Servers and User Mailboxes:**
 - Keep mail server software up to date.
 - Enforce strong password policies and implement multi-factor authentication (MFA).

By considering these threat vectors and implementing corresponding countermeasures, the PR and marketing company can enhance the security of its private network and protect sensitive client information. Regular security training and staying vigilant against emerging threats are essential components of a robust cybersecurity strategy.